The basic principle of Counting

Suppose that there are two experiments are to be performed.

If ① experiment 1 can result in any one of 'm' possible outcomes

and

② for each outcome of experiment 1 there are 'n' possible outcomes of experiment 2

then together there are 'mn' possible outcomes of the two experiments.

Eg: [1] A small community consists of 10 women, each of whom has 3 children. If one women and one of her children are to be choosen as a mother and child of the year, then how many different choices are possible?

Experiment 1 (choosing a mother)

Experiment 2 (choosing her children)

w_1 — 3 children

w_2 — 3 childred.

w_3

$\vdots$

w_{10}

In experiment 1, selecting a women has 10 possible choices (outcomes) and for each women there are three choices in experiment 2.

The total number of possible choices are

$$10 \times 3 = \underline{\underline{30}}$$

For more than two experiments, the basic principle of counting can be generalized.

- Let r experiments are performed. The first one may result in n_1 possible outcomes; and if, for each of these n_1 outcomes there are n_2 possible outcomes of the second experiment and if for each of the possible outcome of the first two experiments, there are n_3 possible outcomes of the third experiment and if, ... (Continues)

then there are $n_1 \cdot n_2 \cdot n_3 \cdots n_r$ possible outcomes of the r experiments.

Eg: $\boxed{1}$: A college planning Committe consists of 3 freshmen, 4 sophomers, 5 juniors and 2 seniors. A Sub Committee of 4, consisting of 1 person from each class is to be choosen. How many different subcommittees are possible?

Sol^n:

The choice of subcommittee can be seen as a combined outcome of four different experiment of choosing single representative from each class.

From the basic principle of counting

there are

$$3 \times 4 \times 5 \times 2 = 120$$

possible sub committees.

Eg 2: How many different 5-place License plates are possible if first two places are to be occupied by letters and final three places by numbers?

Soln: There are 26 letters, for first place there are 26 possible out comes and for each one of them there are 26 possible out comes in second place.

For each choice of first two places there are 10 choices.

similarly, for fourth and fifth place

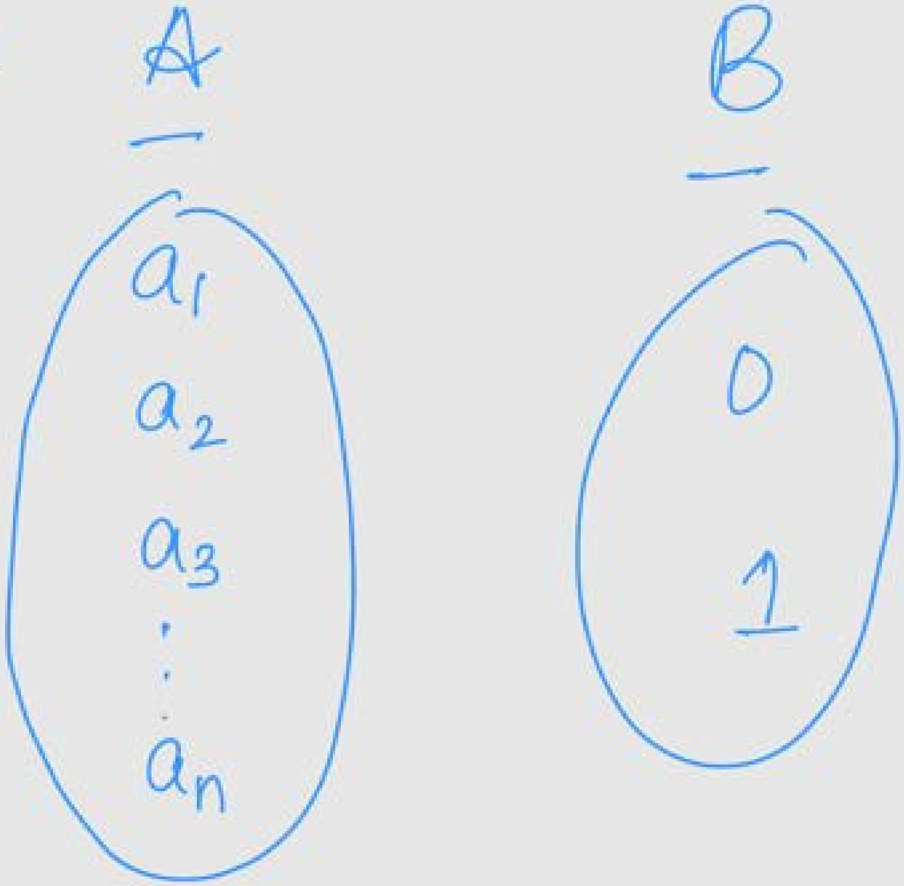
From basic principle of Counting there are

$$26 \times 26 \times 10 \times 10 \times 10$$
$$= 676,000.$$

number of possible license plates.

Eg $\boxed{3}$: Let $A = \{a_1, a_2, a_3, \ldots a_n\}$ and $B = \{0, 1\}$. How many different functions be defined from set A to set B?

Sol^n :



A function $f : A \longrightarrow B$ will

map a_1 to either 0 (or) 1

For $f(a_1)$ there are two choices 0 (or) 1

$f(a_2)$ has two chooses 0 (or) 1

$\vdots$

$f(a_n)$ has two choices 0 (or) 1

Therefore, there are

$$2 \times 2 \times 2 \times \cdots \times 2 \text{ (n times)}$$
$$= 2^n$$

possible functions from A to B.

- In general for any two finite sets A and B, there are $|B|^{|A|}$ number of distinct functions can be defined.

Here $|A|$ = no. of elements in A

$|B|$ = no. of elements in B.

Permutations

Suppose there are 'n' objects.

Each ordered arrangement of 'n' objects is called a "Permutation"

The first object in the permutation can be any one of the 'n' objects.

The second object in the permutation can be choosen from any of the remaining (n-1) objects and so on...

From basic counting principle there are

$$n \cdot (n-1)(n-2)(n-3) \cdots 3 \cdot 2 \cdot 1$$
$$= n!$$

different permutations of the 'n' objects

Eg [1]: How many batting orders are possible for a cricket team consisting of 11 players?

Sol^n: Each batting order refers to a permutation of 11 objects

There are $11!$ possible batting orders.

Eg [2]: How many different letter arrangements can be formed from the letters PEPPER?

Sol^n:- If all P's and E's are distinguished from one another then there are

$6! = 720.$

However there are three P's and two E's, so their permutations result in the same arrangement.

Therefore there are

$$\frac{6!}{3!\,2!} = 60$$

different letter arrangements formed.

Combinations

We are often interested in determining the number of distinct groups of r objects that could be formed from a total number of n objects.

For example: There are 5 objects A, B, C, D, E

To knoco the number of distinct groups of 3 formed from these 5 items:

There are 5 ways of selecting the first item, 4 ways to select next item and 3 ways to sellt the final item. That is there are $5 \times 4 \times 3$ when the order in which they are arranged is relavent.

However, for choosing a group of three item (their) order is irrelavent and each group items arranged 3! ways.

The total number of groups of 3 items formed from 5 items is,

$$\frac{5 \times 4 \times 3}{3 \times 2 \times 1} = 10.$$

In general, the number of distinct groups of r objects formed from 'n' objects is,

$$\frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$$

$$= \frac{n(n-1)(n-2)\cdots(n-r+1)(n-r)!}{r!\,(n-r)!}$$

$$= \frac{n!}{(n-r)!\,r!}$$

For $r \leq n$, the notation used is,

$$\binom{n}{r} = \frac{n!}{(n-r)!\,r!}$$

Eg $\boxed{1}$: A Committee of 3 is to be formed from a group of 20 people. How many different Committee are possible?

Sol^n:- There are $\binom{20}{3}$

$$= \frac{20!}{17!\,3!}$$

$$= 1140$$

possible Committees.

Eg $\boxed{2}$: From a group of 5 women and 7 men, how many different committees consisting of 2 women and 3 men can be formed?

What if 2 men feuding and refuse to serve the committee together?

Soln:- There are $\binom{5}{2}$ possible groups of 2 women and $\binom{7}{3}$ possible groups of 3 men.

From basic Counting principle,

there are $\binom{5}{2} \cdot \binom{7}{3} = 350$

possible Committees Consisting of

2 women and 3 men.

Now suppose 2 of the men

refuse to serve together.

Note that there are

$\binom{2}{2}\binom{5}{1} = 5$ groups

of 3 men contains both of

the feuding men out of

$\binom{7}{3} = 35$ groups of 3 men.

So, there are $35 - 5 = 30$

groups of 3 men that do not

Conting both of the feuding

men.

This follows that

$$30 \times \binom{5}{2} = 30 \times 10$$

$$= 300$$

possible Comittees (such that

both feuding men are not together).

Multinomial Coefficients

In this section we consider the following problem:

" A set of n distinct items is to be divided into r distinct groups of respective sizes $n_1, n_2, n_3, \ldots, n_r$ where $\sum_{i=1}^{r} n_i = n$.

How many different divisions are possible? "

- To answer this, firstly note that there are $\binom{n}{n_1}$ possible choices for group of size n_1.

- After forming first group of size n_1, there are $(n-n_1)$ items remaining.
- For each choice of first group, there are $\binom{n-n_1}{n_2}$ possible choices for the second group.

and so on....

Applying basic counting principle there are

$$\binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-n_1-n_2-\cdots-n_{r-1}}{n_r}$$

$$= \frac{n!}{(n-n_1)!\,n_1!}\,\frac{(n-n_1)!}{(n-n_1-n_2)!\,n_2!}\cdots\frac{(n-n_1-n_2\cdots-n_{r-1})!}{0!\,n_r!}$$

$$= \frac{n!}{n_1!\,n_2!\,n_3!\cdots n_r!}$$

possible divisions.

If $n_1+n_2+n_3+\cdots+n_r=n$, we define $\binom{n}{n_1\,n_2\,n_3\cdots n_r}$ by

$$\binom{n}{n_1\,n_2\,n_3\cdots n_r} = \frac{n!}{n_1!\,n_2!\cdots n_r!}$$

Thus $\binom{n}{n_1\, n_2\, n_3 \cdots n_r}$ represents the number of possible divisions of n distinct objects into r distinct groups of respective sizes $n_1, n_2, \ldots n_r$.

Eg $\boxed{1}$: A police department in a small city consists of 15 officers. If the department policy is to have 10 of the officers patrolling the streets, 2 of the officers working full time at the station and 3 of the officers reserve at the station. In how many ways these officers be deployed?

Solⁿ: The number ways these 15 officers deployed is same as the number possible divisions of 15 into 3 distinct groups of size 10, 2, 3 respectively

The answer is, $\binom{15}{10\ 2\ 3}$

$$= \frac{15!}{10!\ 2!\ 3!}$$

$= 30,030$ possible divisions.

Eg $\boxed{2}$: Ten children are to be divided into A team and a B team of 5 each. How many divisions are possible?

Sol^n:- There are $\dfrac{10!}{5!\,5!}$

$= 252$ possible division.

Problem: The number of integer solutions of Equations.

An individual has gone to the fishing lake that contains r types of fishes namely type 1, type 2, type 3 ... type r.

If we take the result of the fishing trip to be the number of each type of the fish caught.

Let x_i denote the number of type i fish Caught. and a total of n fishes were Caught then

$$x_1 + x_2 + x_3 + \cdots + x_r = n.$$

Thus number of possible out Comes when a total of n fishes were Caught is same as the number of non-negative integer valued $(x_1, x_2, \ldots x_r)$ s.t

$$x_1 + x_2 + \cdots + x_r = n \quad \text{— (1)}$$

First we compute the number of positive integer valued vectors $(x_1, x_2, \ldots, x_r)$ satisfy equation (1).

Suppose there are n fishes lined up there are $(n-1)$ spaces b/w them. We wish to divide them into r distinct groups.

Equivalentely, we need to choose $(r-1)$ spaces from $(n-1)$ spaces

The number of positive integer solution of (1) is

$$\binom{n-1}{r-1}$$

- Now we obtain the number of non-neg. integer solution of

$$x_1 + x_2 + x_3 + \cdots + x_r = n.$$

It is same as the number of positive solutions of equation $y_1 + y_2 + \cdots + y_r = n + r.$

$$(y_i = x_i + 1)$$

Thus,

there are $\binom{n+r-1}{r-1}$ number

of non-negative integer solutions

of equation

$$x_1+x_2+x_3+\cdots+x_r=n$$

Eg $\boxed{3}$: How many distinct non-negative integer-valued solutions of $x_1+x_2=3$ are possible?

Soln:- $n=3, \quad r=2$

$$\binom{n+r-1}{r-1} = \binom{3+2-1}{2-1}$$

$$=4$$